The amino group of proteins can be excreted in three forms in living organisms – ammonia, urea or uric acid. Based on the product of elimination, the living organisms can be classified as ammonotelic, ureotelic and uricotelic respectively. Many terrestrial vertebrates excrete the amino nitrogen as urea, while birds and terrestrial reptiles excrete it in the form of uric acid.

In ureotelic animals, the ammonia that is released in the liver is converted to urea and excreted by the kidneys. The pathway of conversion of ammonia to urea involves ornithine in a cyclic fashion. Hence, the pathway is called as urea cycle or Ornithine cycle or Krebs Henseleit cycle after its discoverers Hans Krebs and a medical student, Kurt Henseleit in 1932.

Urea is produced from ammonia with the help of five enzymes and two of these enzymes are present in the mitochondria, while, the rest are present in the cytosol. Out of the two amino groups of urea, one amino group enters the urea cycle as carbamoyl phosphate, and the other enters as aspartate (Figure 4.11).

The five steps of Urea cycle:

1. **Formation of carbamoyl phosphate:** Carbamoyl phosphate synthetase I present in the mitochondria facilitates the synthesis of carbamoyl phosphate from carbon dioxide, ammonia and ATP. This reaction consumes two ATPs and is irreversible. It is different from the Carbamoyl phosphate synthetase II present in the cytosol, which is involved in pyrimidine biosynthesis, and uses glutamine as the substrate.

2. **Formation of Citrulline:** Carbamoyl phosphate reacts with ornithine to form citrulline and the reaction is catalysed by Ornithine transcarbamoylase. The other three enzymes of urea cycle are present in the cytosol. Hence, Citrulline is transported out of the mitochondria by a transporter.

3. **Formation of argininosuccinate:** Citrulline and aspartate condense together to form argininosuccinate in the presence of the enzyme Argininosuccinate synthetase. This reaction requires ATP and is catalyzed by a citrullyl-AMP intermediate (the amino group of Aspartate provides the second nitrogen atom for Urea synthesis).

4. **Formation of arginine from argininosuccinate:** This reaction catalysed by argininosuccinase releases arginine and fumarate (which enters the citric acid cycle).

5. **Formation of urea and regeneration of ornithine:** The final step in urea synthesis is the release of urea from arginine by the enzyme arginase and this step regenerates ornithine. The ornithine thus formed enters into the mitochondria with the help of a transporter.

One molecule of Urea is synthesized from one molecule of carbondioxide, one molecule of ammonia and the amino group of aspartate. The reactions consume four high energy bonds equivalent to four ATPs.

XII U4 Protein.indd 102 4/8/2020 3:09:24 PM